

ApproxDA-TransUNet: Understanding Window Sensitivity of Attention Approximation for Medical Image Segmentation

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Abstract—Dual attention, which combines spatial and channel self-attention, is effective for medical image segmentation, but its quadratic cost has motivated windowed, low-rank, and grouped approximations that are usually treated as accuracy-neutral. How these approximations affect segmentation accuracy, and whether the effect is the same across tasks, has received little controlled study. We present ApproxDA-TransUNet, which keeps the architecture of DA-TransUNet and replaces only its attention modules with an approximation controlled by three independent parameters: spatial window size M , projection rank r , and channel group number G . Varying one parameter at a time on Synapse multi-organ CT, Kvasir-SEG, and ISIC 2018, we observe strongly dataset-dependent window sensitivity: the Dice range across window sizes is **2.30** points on Synapse but only **0.64** and **0.50** points on Kvasir-SEG and ISIC 2018. With an appropriately selected window, ApproxDA-TransUNet improves Dice over a DA-TransUNet re-trained under the same protocol on all three datasets, while using fewer parameters and FLOPs in its attention blocks. These results indicate that attention approximation is not accuracy-neutral and that its window scale should be treated as a task-dependent design choice.

Index Terms—medical image segmentation, dual attention, attention approximation, window size, ablation study

I. INTRODUCTION

Transformers have advanced medical image segmentation by modeling long-range dependencies that convolutional networks capture only indirectly. Dual attention [1]—a Position Attention Module (PAM) for spatial correlations and a Channel Attention Module (CAM) for channel correlations—is one such mechanism, and DA-TransUNet [2] showed its effectiveness in a hybrid CNN–Transformer architecture for multi-organ and polyp segmentation. Full PAM costs $\mathcal{O}(N^2C)$ for N positions and C channels, and windowed attention [3], low-rank projection [4], and grouped channel interaction [5] are standard ways to reduce this cost. These approximations are usually adopted as accuracy-neutral engineering choices, which raises a question that is rarely examined across tasks with different spatial characteristics:

Does attention approximation change segmentation accuracy, and does the effect depend on the task?

To study this question we design **ApproxDA-TransUNet**¹, which keeps the architecture of DA-TransUNet and replaces only its attention modules with approximations controlled by three independent parameters. With the backbone, training protocol, and evaluation fixed, we vary one parameter at a time on three datasets with different anatomical and visual characteristics, and compare against DA-TransUNet re-trained under the same protocol.

The main contributions of this paper are:

- **A controlled dual-attention approximation framework.** Window size M , rank r , and group number G control the spatial scope, the affinity resolution, and the channel-interaction scope of dual attention. In its exact limit the block reduces to the DA-TransUNet block, so each approximation can be ablated in isolation.
- **Evidence of dataset-dependent window sensitivity.** Synapse multi-organ CT is substantially more sensitive to M than Kvasir-SEG and ISIC 2018, and with an appropriately selected window ApproxDA-TransUNet improves Dice over the matched DA-TransUNet on all three datasets.

The remainder of this paper is organized as follows: Section II reviews related work; Section III presents ApproxDA-TransUNet; Section IV describes the protocol and main results; Section V reports the controlled ablations; Section VI concludes.

II. RELATED WORK

A. CNN-based Medical Image Segmentation

Convolutional encoder-decoder architectures established the foundational paradigm for medical image segmentation, with U-Net [6] introducing the skip-connection design that underpins most segmentation networks, and successors [7]–[10] progressively adding dense connections, attention gating, and residual learning for improved feature reuse and training stability. These models offer strong computational efficiency and favorable inductive biases for local texture patterns, but their

¹Source code will be made available upon acceptance.

limited receptive fields make it difficult to model relationships between distant anatomical structures.

B. Transformer-based Medical Image Segmentation

Transformer architectures have significantly advanced medical image segmentation by enabling global dependency modeling beyond the locality of convolutional neural networks. Representative methods such as TransUNet [11], Swin-Unet [12], UNETR [13], UCTransNet [14], TransNorm [15], and DAE-former [16] combine convolutional feature extraction with transformer-based contextual modeling and report strong segmentation performance across a wide range of medical imaging tasks. These methods typically apply the same attention design across tasks, and how the spatial scope of attention affects accuracy on different tasks is rarely examined.

C. Dual Attention and DA-TransUNet

DANet [1] introduced Position Attention Modules (PAM) and Channel Attention Modules (CAM), capturing spatial and channel correlations at $\mathcal{O}(N^2C)$ and $\mathcal{O}(C^2N)$ complexity respectively. DA-TransUNet [2] adapted this design for medical image segmentation by inserting dual-attention blocks at multiple skip connections of a hybrid R50-ViT-B/16 encoder-decoder, improving accuracy while inheriting the quadratic cost of full-map attention. DBAANet [17] further extends dual-branch attention by aggregating position and channel attention across multiple scales, demonstrating the continued appeal of dual-attention paradigms for medical image segmentation.

D. Efficient Attention and Adaptive Routing

Windowed local attention [3], low-rank projection [4], and grouped channel interaction [5] have become standard strategies for reducing the cost of global attention, with learnable gating widely adopted to fuse multi-branch outputs [5], [16]. DAMamba [18] further shows that dual-attention routing transfers effectively to state-space models for lightweight breast ultrasound segmentation, confirming the versatility of multi-branch attention beyond standard transformer architectures. Existing efficient attention methods are mainly evaluated by their computational savings, and their effect on accuracy is usually assumed to be small. Whether this holds equally across segmentation tasks with different spatial characteristics has received little controlled study. This work addresses that question by varying one approximation parameter at a time within a fixed architecture and training protocol.

III. METHOD

A. Architecture overview

ApproxDA-TransUNet keeps the hybrid CNN-ViT encoder and decoder of DA-TransUNet [2], with an R50-ViT-B/16 backbone (Fig. 1). Three CNN stages produce features at strides 2, 4, and 8, and a 12-layer ViT processes 14×14 patch tokens. ApproxDABlocks are placed at the bottleneck and at the three skip connections (512 channels at 28^2 , 256 at 56^2 , 64 at 112^2). The public DA-TransUNet implementation applies

dual attention only at the bottleneck and the 512- and 256-channel skips, because its decoder selects the block by the decoder channel count rather than the skip channel count; we keep that implementation unchanged as the baseline and additionally place a block at the 64-channel skip.

Each ApproxDABlock keeps the layout of the DA-TransUNet block and replaces only its attention modules (Fig. 2). Each branch first reduces the C input channels to $C/16$ with a 3×3 convolution, batch normalization, and ReLU, as in DA-TransUNet; the PAM and CAM branches then apply the approximated attention described below, followed by another 3×3 convolution block. The two branches are fused and projected back to C channels.

Approximation axes. The original dual-attention block computes dense spatial and channel attention over the full feature map. ApproxDA-TransUNet approximates it along three independent axes. *Spatial scope* (window size M) controls how far PAM reaches; $M \times M$ windows trade long-range context for lower cost. *Affinity resolution* (rank r) controls how finely attention is computed inside a window; higher r approaches exact attention. *Channel interaction* (groups G) restricts CAM to interactions within G channel groups. Any two of $\{M, r, G\}$ can be held fixed while the third varies.

B. Spatial approximation

Full PAM. Following [1], let $\mathbf{A} \in \mathbb{R}^{C \times N}$ be the (channel-reduced) input of the attention module, $N = HW$, and let $\mathbf{B}, \mathbf{C} \in \mathbb{R}^{C' \times N}$ and $\mathbf{D} \in \mathbb{R}^{C \times N}$ be query, key, and value features produced by 1×1 convolutions, with columns $\mathbf{B}_j$, $\mathbf{C}_i$, $\mathbf{D}_i$ and $C' = \max(1, C/8)$ as in the DA-TransUNet implementation. PAM computes

$$s_{ji} = \frac{\exp(\mathbf{B}_j^\top \mathbf{C}_i)}{\sum_{i'=1}^N \exp(\mathbf{B}_j^\top \mathbf{C}_{i'})}, \quad \mathbf{E}_j = \alpha \sum_{i=1}^N s_{ji} \mathbf{D}_i + \mathbf{A}_j, \quad (1)$$

where α is a learned scalar initialized to 0. The $N \times N$ attention matrix costs $\mathcal{O}(N^2C)$.

Window partitioning. Following [3], the map is split into non-overlapping $M \times M$ windows with $M \leftarrow \min(M, H, W)$, and Eq. (1) is evaluated inside each window over its $N_w = M^2$ positions. This reduces the cost to $\mathcal{O}(NM^2C)$; the window outputs are then merged back to the $H \times W$ grid.

Low-rank projection. Following [4], keys and values inside each window, $\mathbf{C} \in \mathbb{R}^{C' \times N_w}$ and $\mathbf{D} \in \mathbb{R}^{C \times N_w}$, are compressed along the token axis by a learned projection $\mathbf{P} \in \mathbb{R}^{N_w \times r}$ shared across windows:

$$\tilde{\mathbf{C}} = \mathbf{C}\mathbf{P} \in \mathbb{R}^{C' \times r}, \quad \tilde{\mathbf{D}} = \mathbf{D}\mathbf{P} \in \mathbb{R}^{C \times r}. \quad (2)$$

Each query then attends to the r projected keys instead of the N_w original ones:

$$\tilde{s}_{jk} = \frac{\exp(\mathbf{B}_j^\top \tilde{\mathbf{C}}_k)}{\sum_{k'=1}^r \exp(\mathbf{B}_j^\top \tilde{\mathbf{C}}_{k'})}, \quad \mathbf{E}_j = \alpha \sum_{k=1}^r \tilde{s}_{jk} \tilde{\mathbf{D}}_k + \mathbf{A}_j, \quad (3)$$

so the per-window attention matrix is $N_w \times r$ instead of $N_w \times N_w$, and the total cost becomes $\mathcal{O}(NrC)$. As in [1], no temperature scaling is applied. When M is clamped to a

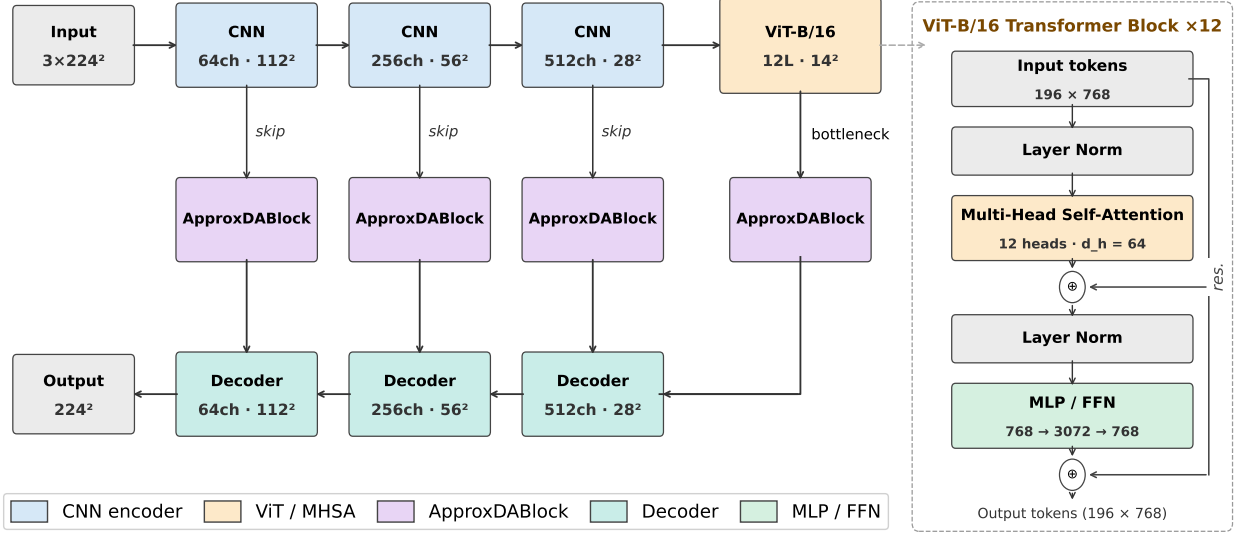


Fig. 1. ApproxDA-TransUNet. ApproxDABlocks (windowed low-rank PAM + grouped CAM + routing) are placed at all three skip connections and at the bottleneck; the CNN-ViT backbone of DA-TransUNet is unchanged.

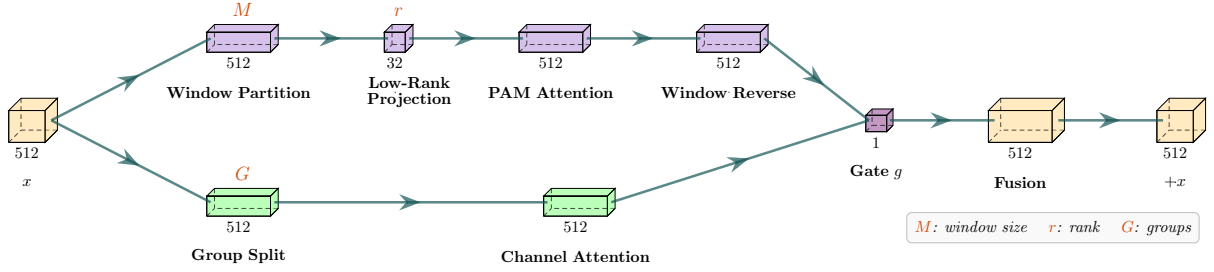


Fig. 2. ApproxDABlock with three independently controllable approximation axes: spatial scope (M), affinity resolution (r), and channel interaction (G). Each branch reduces the input to $C/16$ channels before attention, as in DA-TransUNet; the branch outputs are fused by the per-channel routing vector $\mathbf{g}$ and projected back to C channels by a 1×1 convolution.

smaller map, the first N_w rows of $\mathbf{P}$ are used. Hence $M=112$ gives a single window per map (global spatial scope), but the attention remains rank- r and is *not* exact full PAM; omitting the projection ($\tilde{\mathbf{C}}=\mathbf{C}$, $\tilde{\mathbf{D}}=\mathbf{D}$) recovers exact attention within each window.

C. Channel approximation

CAM [1] computes a $C \times C$ channel affinity at $\mathcal{O}(C^2N)$ cost. We split the C channels into G groups of $C_g = C/G$ channels and compute attention within each group. For group g with channel rows $\mathbf{A}_1^{(g)}, \dots, \mathbf{A}_{C_g}^{(g)} \in \mathbb{R}^N$,

$$x_{ij}^{(g)} = \frac{\exp(-\mathbf{A}_i^{(g)} \cdot \mathbf{A}_j^{(g)})}{\sum_{k=1}^{C_g} \exp(-\mathbf{A}_i^{(g)} \cdot \mathbf{A}_k^{(g)})}, \quad (4)$$

$$\mathbf{E}_i^{(g)} = \beta \sum_{j=1}^{C_g} x_{ij}^{(g)} \mathbf{A}_j^{(g)} + \mathbf{A}_i^{(g)},$$

where β is a learned scalar initialized to 0. Following the official DANet and DA-TransUNet implementations, the chan-

nel affinity is negated before the softmax (implemented as $\max_k \mathbf{A}_i \cdot \mathbf{A}_k - \mathbf{A}_i \cdot \mathbf{A}_j$, which is equivalent under the softmax). Grouping reduces the cost to $\mathcal{O}(C^2N/G)$, and $G=1$ recovers the CAM module of DA-TransUNet exactly. When a reduced map has fewer than G channels (4 channels at the 64-channel skip), one channel per group is used.

D. Routing and complexity

Let $\mathbf{X}$ be the block input with C channels, and let ϕ denote a 3×3 convolution with batch normalization and ReLU. The block output is

$$\begin{aligned} \mathbf{P} &= \phi_{51}(\text{PAM}(\phi_{5a}(\mathbf{X}))), & \mathbf{Q} &= \phi_{52}(\text{CAM}(\phi_{5c}(\mathbf{X}))), \\ \mathbf{Y} &= \text{ReLU}(\mathbf{W}_f(\mathbf{g} \odot \mathbf{P} + (\mathbf{1} - \mathbf{g}) \odot \mathbf{Q})), \end{aligned} \quad (5)$$

where ϕ_{5a}, ϕ_{5c} map $C \rightarrow C/16$ channels, $\mathbf{W}_f$ is a 1×1 convolution back to C channels (with dropout during training), and $\mathbf{g} \in [0, 1]^{C/16}$ is a per-channel routing vector broadcast over positions, set by the routing mode:

PAM-only: $g = 1$
 CAM-only: $g = 0$
 fixed: $g = 0.5 \cdot 1$
 learned: $g = \sigma(\mathbf{W}_g \text{ GAP}(\mathbf{X}))$

PAM-only and CAM-only routing build only the branch they use. With exact attention (M covering the whole map, no projection), $G=1$, and $g=0.5 \cdot 1$, the block reduces to the DA-TransUNet block up to a rescaling of $\mathbf{W}_f$. The network is trained with $\mathcal{L} = \frac{1}{2}\mathcal{L}_{\text{Dice}} + \frac{1}{2}\mathcal{L}_{\text{CE}}$, as in DA-TransUNet.

Windowing and low-rank projection reduce the PAM attention matrix from $N \times N$ to $N \times r$, a theoretical reduction of the attention-matrix term by $N/r \approx 390$ at the 112^2 skip with $r=32$. This applies to the attention operation only; the measured cost of the attention blocks and of the whole network is reported in Section IV-C.

IV. EXPERIMENTS

A. Datasets and Protocol

Datasets. *Synapse* [19]: 30 abdominal CT scans covering 8 organs, with the standard 18/12 train/test split. *Kvasir-SEG* [20]: 1000 colonoscopy images with binary polyp masks, using the 880/120 train/test split recommended with the dataset. *ISIC 2018* [21]: dermoscopy images with binary lesion masks, using the official Task 1 training (2594), validation (100), and test (1000) sets.

Model selection. No checkpoint selection is performed: every model, including the re-trained DA-TransUNet, is evaluated at its final training epoch, as in TransUNet [11]. On ISIC 2018, M , r , G , and routing are selected on the official validation set. Synapse and Kvasir-SEG have no validation split; following common practice on Synapse [2], [11], their configuration is taken from the ablations in Section V, which report every configuration on the test set (see the limitations in Section V-C). Each test set is otherwise used only for final evaluation.

Training. All models are trained for 300 epochs with ImageNet-21K pretrained R50-ViT-B/16 weights, 224×224 input, random rotation and flipping, batch size 24, polynomial learning-rate decay, and loss $\frac{1}{2}\mathcal{L}_{\text{CE}} + \frac{1}{2}\mathcal{L}_{\text{Dice}}$, on the same hardware (one NVIDIA Tesla T4). Following DA-TransUNet [2], Synapse uses SGD (learning rate 0.01, momentum 0.9, weight decay 10^{-4}), and Kvasir-SEG and ISIC 2018 use Adam (learning rate 10^{-3} , weight decay 10^{-4}). We train longer than the original protocols (150 epochs on Synapse; 500 and 50 epochs on Kvasir-SEG and ISIC 2018) and use 224×224 rather than 256×256 input on all datasets, so that the same window sizes M apply everywhere. DA-TransUNet is re-trained with exactly the same data, splits, and protocol. Unless stated otherwise, ApproxDA-TransUNet uses $r=32$ and PAM-only routing; G ($= 8$ by default) only matters for routings that use CAM.

Metrics. Synapse: mean Dice (DSC) and 95% Hausdorff distance (HD95) over the 8 organs, computed with the TransUNet evaluation code. Kvasir-SEG and ISIC 2018: DSC and mean IoU (mIoU); we do not report HD95

TABLE I SEGMENTATION RESULTS. ROWS MARKED $\dagger$ ARE VALUES REPORTED IN THE LITERATURE UNDER THEIR ORIGINAL PROTOCOLS AND ARE SHOWN FOR REFERENCE ONLY; THE MATCHED COMPARISON IS BETWEEN THE LAST TWO ROWS.

Method	Synapse		Kvasir-SEG		ISIC 2018	
	DSC $\uparrow$	HD95 $\downarrow$	DSC $\uparrow$	mIoU $\uparrow$	DSC $\uparrow$	mIoU $\uparrow$
U-Net $\dagger$ [6]	76.85	39.70	88.22	80.12	87.22	81.14
U-Net++ $\dagger$ [7]	76.91	36.93	86.57	77.67	87.30	81.33
Res-UNet $\dagger$ [9]	76.95	38.44	77.85	66.04	83.32	76.51
MultiResUNet $\dagger$ [10]	77.42	36.84	—	—	—	—
Att-UNet $\dagger$ [8]	77.77	36.02	86.61	78.01	87.60	81.51
TransUNet $\dagger$ [11]	77.48	31.69	87.91	80.03	88.78	82.63
UCTransNet $\dagger$ [14]	78.23	26.75	—	—	—	—
TransNorm $\dagger$ [15]	78.40	30.25	—	—	—	—
MIM $\dagger$ [22]	78.59	26.59	—	—	—	—
Swin-UNet $\dagger$ [12]	79.13	21.55	—	—	—	—
DA-TransUNet $\dagger$ [2]	79.80	23.48	88.47	—	88.88	82.78
DA-TransUNet (re-trained)	—	—	88.44	81.70	88.43	80.68
ApproxDA-TransUNet (ours)	80.94	27.49	90.17	84.27	89.55	—

ApproxDA-TransUNet: PAM-only routing, $r=32$; M = Synapse 28, Kvasir-SEG 56, ISIC 2018 28, selected from the ablations (Synapse, Kvasir-SEG) or on the validation set (ISIC 2018). Literature rows use different splits and training protocols (e.g., DA-TransUNet uses 256×256 input and Adam for 500/50 epochs on Kvasir-SEG/ISIC 2018).

for these datasets because their pixel spacing is not calibrated. [TODO: main comparison: mean $\pm$ std over 3 seeds; Synapse per-organ DSC; paired test / bootstrap CI (experiments/statistical_validation.py).]

B. Main Results

Table I reports the results. Under matched training and evaluation, ApproxDA-TransUNet improves DSC over the re-trained DA-TransUNet on all three datasets (+1.73 on Kvasir-SEG, +1.12 on ISIC 2018, [TODO: Synapse after re-run]). The gains are metric-specific rather than uniform: Synapse HD95 is worse than the DA-TransUNet and Swin-UNet values reported in the literature, and ISIC 2018 mIoU is below the reported DA-TransUNet value. Literature rows are included for context only and are not used to support our claims. Fig. 3 shows representative predictions.

C. Computational Cost

Table II compares parameters and FLOPs, counted with the same tool and input for all models. The unchanged backbone dominates end-to-end cost: the ViT encoder alone holds 85.1M parameters and 17.4 GFLOPs, and the CNN encoder and decoder most of the rest. The approximation reduces the cost of the attention blocks from 1.37 GFLOPs (exact attention) to 0.40 GFLOPs, below the 0.87 GFLOPs of the DA-TransUNet attention blocks even though ApproxDA-TransUNet uses one additional block at the 112^2 skip. At the whole-network level the saving is small (29.73 vs. 30.20 GFLOPs), so we do not claim whole-network acceleration.

V. CONTROLLED ABLATIONS

Each ablation varies one approximation parameter while all others, the backbone, and the training protocol are fixed.

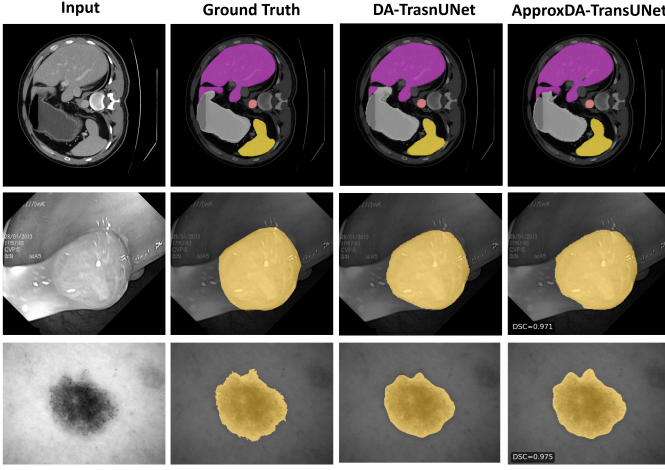


Fig. 3. Qualitative comparison (columns: input, ground truth, DA-TransUNet, ApproxDA-TransUNet). Top: Synapse ($M=28$); middle: Kvasir-SEG ($M=56$); bottom: ISIC 2018 ($M=28$).

TABLE II PARAMETERS AND FLOPs (FVCORE, ONE $1 \times 224 \times 224$ SLICE; ONE MULTIPLY-ADD COUNTED AS ONE FLOP). “ATTN.” IS THE GFLOPs OF THE DUAL-ATTENTION BLOCKS THAT ARE EXECUTED.

Model	Params (M)	GFLOPs	Attn.
DA-TransUNet [2]*	107.95	30.20	0.87
ApproxDA, exact ($M=112$, no proj.)	105.89	30.71	1.37
ApproxDA ($M=28$, $r=32$, PAM)	105.99	29.73	0.40
ApproxDA ($M=7$, $r=32$, learned)	106.51	30.08	0.74

*Public implementation; 1.40M of its parameters belong to attention blocks that are constructed but never executed. DA-TransUNet applies attention at three locations, ApproxDA-TransUNet at four.

A. Window Size and Empirical Window Sensitivity

Table III and Fig. 4 vary only M . On Synapse, DSC peaks at an intermediate window ($M=28$) and drops for both smaller and larger windows, giving a 2.30-point range. Kvasir-SEG and ISIC 2018 are 3.6–4.6 $\times$ less sensitive, with ranges of 0.64 and 0.50 points. We summarize this behavior by the *empirical window sensitivity*

$$S_{\text{window}} = \max_M \text{DSC}(M) - \min_M \text{DSC}(M),$$

which describes how this architecture responds to M on a given dataset; it is not an independent predictor of a task’s contextual requirement, and three datasets are not sufficient to establish it as a task property. The observed difference is consistent with multi-organ CT segmentation depending more on inter-organ spatial context than polyp and skin-lesion segmentation, which are largely determined by local appearance and boundary contrast.

The sensitivity also bounds how much the choice of M matters in practice: on Kvasir-SEG and ISIC 2018 any tested window stays within 0.64 points of the best, whereas on Synapse the window must be tuned. On all three datasets the best window **exceeds** the re-trained DA-TransUNet, suggesting a locality-induced bias. This comparison changes both the attention approximation and the routing (PAM-only instead of PAM + CAM); the exact-attention rows of Table IV isolate

TABLE III DSC (%) VERSUS WINDOW SIZE M (PAM-ONLY ROUTING, $r=32$). RANGE = $\max_M \text{DSC}(M) - \min_M \text{DSC}(M)$.

Setting	Synapse	Kvasir-SEG	ISIC 2018
$M=7$	78.64	89.53	89.41
$M=28$	80.94	89.54	89.55
$M=56$	78.90	90.17	89.05
$M=112$ (global, low-rank)	79.44	89.56	89.51
Range	2.30	0.64	0.50
DA-TransUNet (re-trained)	–	88.44	88.43

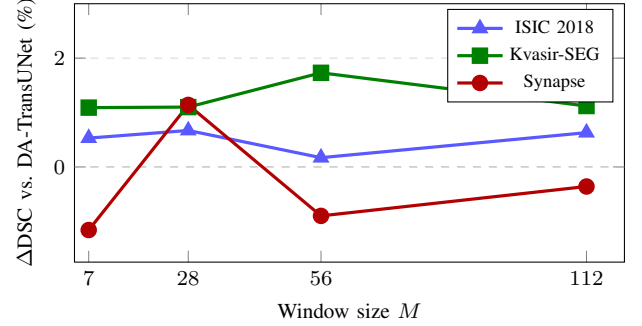


Fig. 4. DSC difference to DA-TransUNet versus window size M (PAM-only routing, $r=32$); raw values in Table III. **Curves from the BIBM runs; to be regenerated from the re-trained models.**

the approximation itself. Note that $M=112$ keeps the rank-32 projection and therefore differs from full PAM.

Fig. 5 visualizes the effect of the window on Kvasir-SEG. On 10 test images, windowed PAM ($M=7$) places 62.3% of its contribution inside the polyp region, compared with 34.2% for global PAM ($M=112$), and the windowed share is higher in 9 of the 10 images. Windowed attention concentrates on polyp boundaries, whereas global attention spreads over surrounding tissue. This observation is consistent with the locality interpretation but does not by itself establish a causal link to accuracy. **Maps from the BIBM checkpoints; to be regenerated.**

B. Rank, Grouping, and Routing

Table IV varies the remaining factors one at a time. Raising r from 32 to 64 does **not** improve DSC, so in the tested range the window scale matters more than the rank. The no-projection rows use exact attention: at $M=112$ this is exact global PAM, which isolates the effect of the approximation from that of the routing, and at $M=28$ it removes the projection at the best Synapse window. G is ablated under CAM-only routing, since the CAM branch does not contribute under PAM-only routing. **[TODO: grouping result.]**

Among the routing modes, PAM-only routing performs best. The learned gate converged close to $g=0.5$ in our Synapse experiments and did not outperform PAM-only routing; we report this as an empirical observation and use PAM-only routing in the other experiments. **[TODO: optional: per-channel gate distribution / trajectory of the learned gate.]**

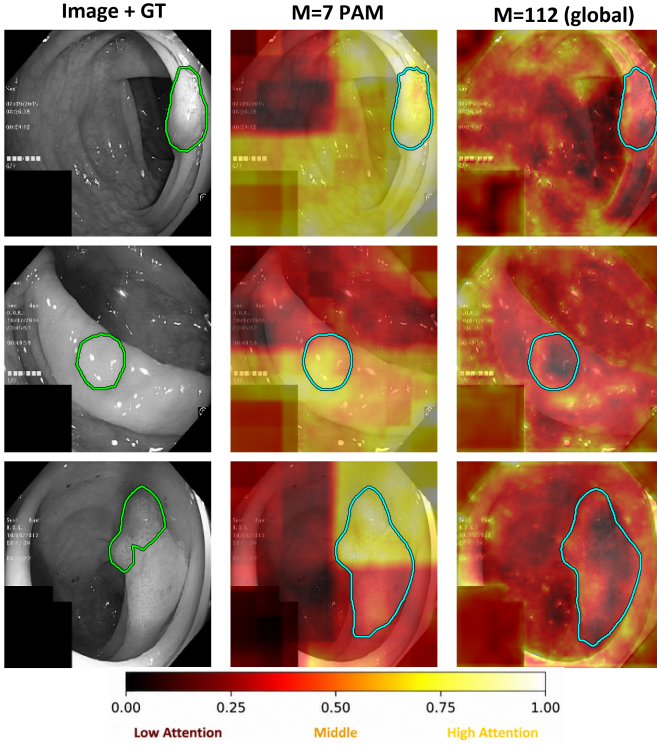


Fig. 5. PAM contribution maps ($\|output - input\|_2$) on three Kvasir-SEG images. Each row: input with ground-truth contour (left), windowed PAM with $M=7$ (center), global PAM with $M=112$ (right). Each panel is normalized independently, so magnitudes are not comparable across panels.

TABLE IV SINGLE-FACTOR ABLATIONS ON SYNAPSE. EACH BLOCK VARIES ONE FACTOR; THE OTHERS ARE FIXED AS INDICATED.

Factor (fixed)	Setting	DSC $\uparrow$	HD95 $\downarrow$
Rank r ($M=112$, PAM)	$r=16$	—	—
	$r=32$	79.44	27.26
	$r=64$	78.93	31.21
	no projection	—	—
Window only ($M=28$, PAM)	no projection	—	—
Groups G (CAM-only)	$G=1$	—	—
	$G=4$	—	—
	$G=8$	78.26	30.59
Routing ($M=7$, $r=32$)	PAM-only	78.64	31.09
	CAM-only	78.26	30.59
	fixed $g=0.5$	—	—
	learned	77.78	34.29

C. Limitations

Because Synapse and Kvasir-SEG have no validation split, their configurations were selected from the test-set ablations above, so the reported ApproxDA-TransUNet results on these datasets may be optimistic. The bias is bounded by the window-sensitivity range: small on Kvasir-SEG (0.64 points), but up to 2.30 points on Synapse, where fine window selection is precisely what our results call for; in practice, M should be tuned on held-out data. The study covers three datasets and a single backbone, and whether window sensitivity can

be predicted without window ablations remains open.

VI. CONCLUSION

This paper presented ApproxDA-TransUNet, a controlled approximation of dual attention that keeps the DA-TransUNet architecture and varies window size, projection rank, and channel grouping one at a time. Across Synapse, Kvasir-SEG, and ISIC 2018, the effect of the window size is strongly dataset-dependent: large on multi-organ CT and small on polyp and skin-lesion segmentation. With an appropriately selected window, ApproxDA-TransUNet improves Dice over DA-TransUNet re-trained under the same protocol, and its attention blocks require fewer FLOPs, although the end-to-end cost remains dominated by the backbone. These results suggest that attention approximation should be treated as a task-dependent design choice rather than an accuracy-neutral efficiency measure. Predicting the appropriate window without window ablations, and testing whether the observed sensitivity holds for other backbones and datasets, are left for future work.

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